

Manoj Sai Penugonda

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SUMMARY

Data Scientist with experience in predictive modeling, Generative AI, and prompt engineering. Skilled in building advanced ML Solutions and transforming raw data into actionable insights that improve business outcomes.

EDUCATION

Anna University
Madras Institute of Technology

08/2018 - 07/2022
B.E. in Electronics Engineering

EXPERIENCE

Data Scientist
Accenture

10/2022 - Present
Chennai

Data Profiling

- Profiled over 5000 records to identify and resolve 1500+ data quality issues, improving **data accuracy** by **15%**.
- Performed comprehensive **data profiling** on multiple datasets, detecting inconsistencies and discrepancies, and providing **actionable insights** that reduced data cleaning time by **20%**.
- Analyzed and documented **key data patterns**, enabling better data organization and improving data usability by **25%**.

ML Based Synthetic Data Generation Bot

- Engineered an ML-driven bot that generated over 1000+ rows of data while maintaining 95% fidelity to the original dataset's **statistical relationship, correlation, and distributions**.
- Built a **regression-based ML model** that achieved **80% accuracy** in predicting weekly product sales volume based on promotional data and historical trends.
- Integrated the synthetic data generation pipeline with downstream analytics tools, accelerating model training by **25%** and **reducing overfitting** by **20%**.
- Designed and implemented a data augmentation pipeline using statistical sampling and generative models, increasing dataset diversity by **30%** and improving **model generalization** on unseen data.

Text Similarity Search - RAG Framework

- Led a **GenAI PoC** for the conversion of user stories into test cases, increasing the project pipeline efficiency by **30%**.
- Executed semantic similarity search on historical text data, comparing user input against past records and retrieving **top-K** matches with **95%** precision.
- Developed a retrieval-optimized chatbot leveraging **Hugging Face Embedding Model** (thenlper/gte-small) and **ChromaDB**, to improve search accuracy by over **25%** using **cosine similarity** techniques.
- Optimized vector search efficiency by **20%** through refined similarity search parameters and improved data preprocessing, enhancing retrieval speed and relevance for large-scale datasets.
- Created a user-friendly GUI with **Streamlit**, reducing decision-making time for non-technical stakeholders by over **40%** and significantly improving accessibility.

TECHNICAL SKILLS

Programming and Query Languages: Python, SQL, React.js

Machine Learning: Optimization Techniques, ML Optimizers, Linear & Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, AdaBoost, Neural Networks (ANN,CNN,RNN,LSTM), Generative AI (LLM Fine-Tuning -LoRA, GANs), Diffusion Models, NLP, Clustering

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow

Tools & Technologies: Azure, Git, GitHub, RESTful APIs, Node.js

Analytics: Statistical Analysis, Data Visualization, Data Wrangling, Feature Engineering, Hypothesis Testing

CERTIFICATIONS

- AZ-500 Microsoft Azure Security Engineer Associate.
- Certificate Programme in Data Science and Machine Learning — IIT Delhi.